COVID-19 Global Impact Dashboard

Detailed Analytics Report

A complete portfolio-style project for analyzing global COVID-19 cases, deaths and vaccination rollout with population normalization and time-series metrics.

1. Project Overview

The project follows a practical analytics pipeline: acquire public data, clean and validate it, derive decision-friendly metrics, visualize trends, and translate the results into plain language. The analysis is designed for a non-technical audience while keeping the computational logic reproducible.

Project question

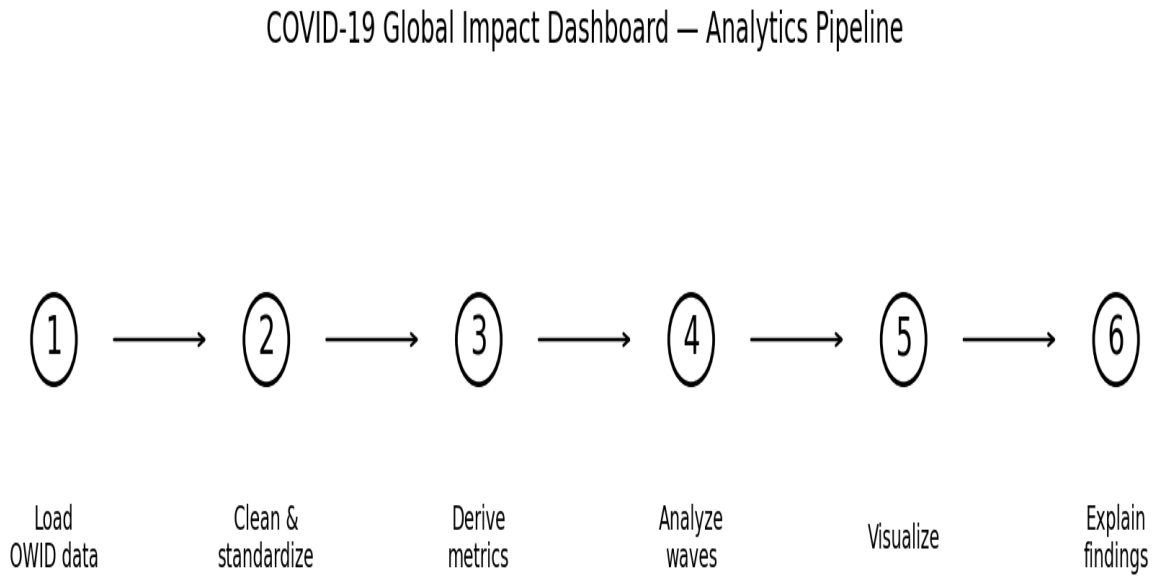
How did COVID-19 waves and vaccination rollout differ across countries, and what patterns become visible when cases and deaths are normalized by population?

Secondary question

Did higher vaccination coverage coexist with lower cumulative confirmed-death rates? This is investigated as an exploratory association only; the dashboard does not claim a causal effect.

Deliverables

Notebook, cleaned/exportable dataset, preview charts, interactive HTML dashboard, one-page executive summary, and this detailed report.



2. Data Source and Scope

The project uses Our World in Data (OWID) COVID-19 data. OWID's current technical documentation describes a compact COVID-19 dataset that consolidates relevant indicators and also provides separate datasets for cases/deaths and vaccinations.

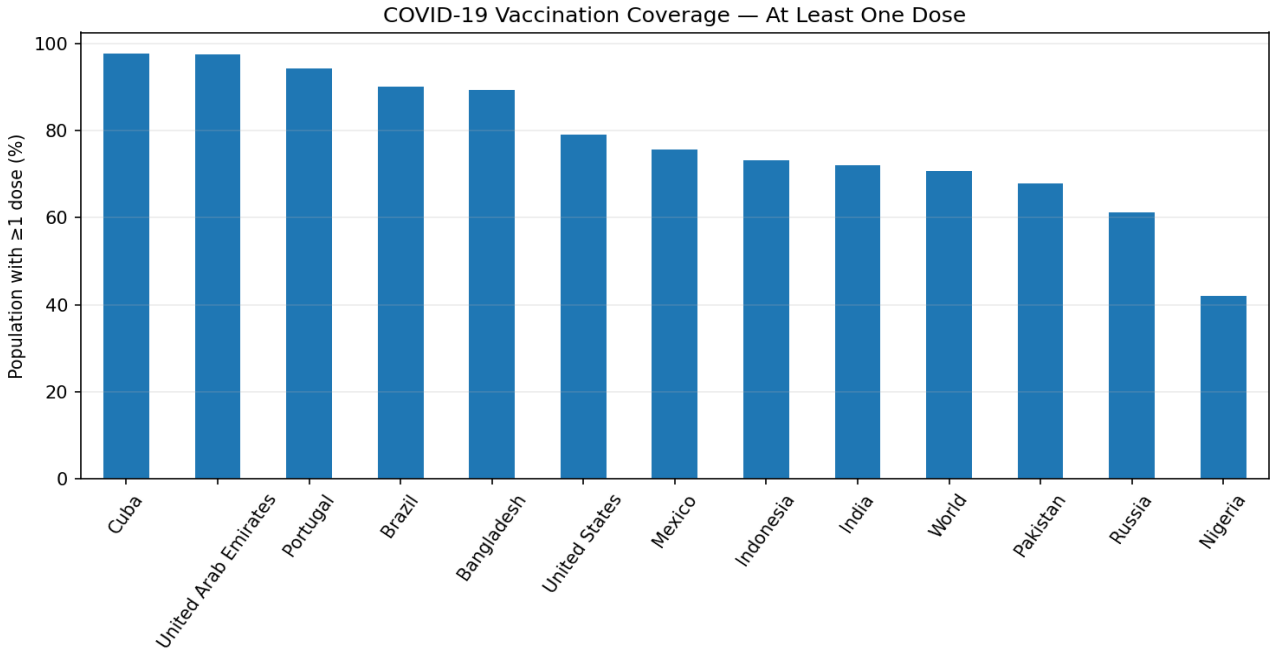
Source fields used

Field	Purpose
location / date	Country-level time-series key.
population	Denominator for per-million metrics.
new_cases / total_cases	Daily and cumulative case analysis.
new_deaths / total_deaths	Daily and cumulative death analysis.
people_vaccinated	At-least-one-dose rollout.
people_fully_vaccinated	Completed initial vaccination protocol.

OWID explains that confirmed cases and deaths are subject to differences in testing, reporting and cause-of-death classification. For that reason, this project avoids treating confirmed figures as the exact number of infections or deaths caused by COVID-19.

Time-period note

The static vaccination preview is an OWID snapshot dated 12 Aug 2024. The included death-per-million preview uses a separate current OWID chart context dated 19 Jul 2026. These dates are deliberately labeled and are not treated as a single synchronized statistical snapshot. The notebook downloads the full current time series for the actual analysis.



3. Data Cleaning and Quality Controls

Cleaning rules

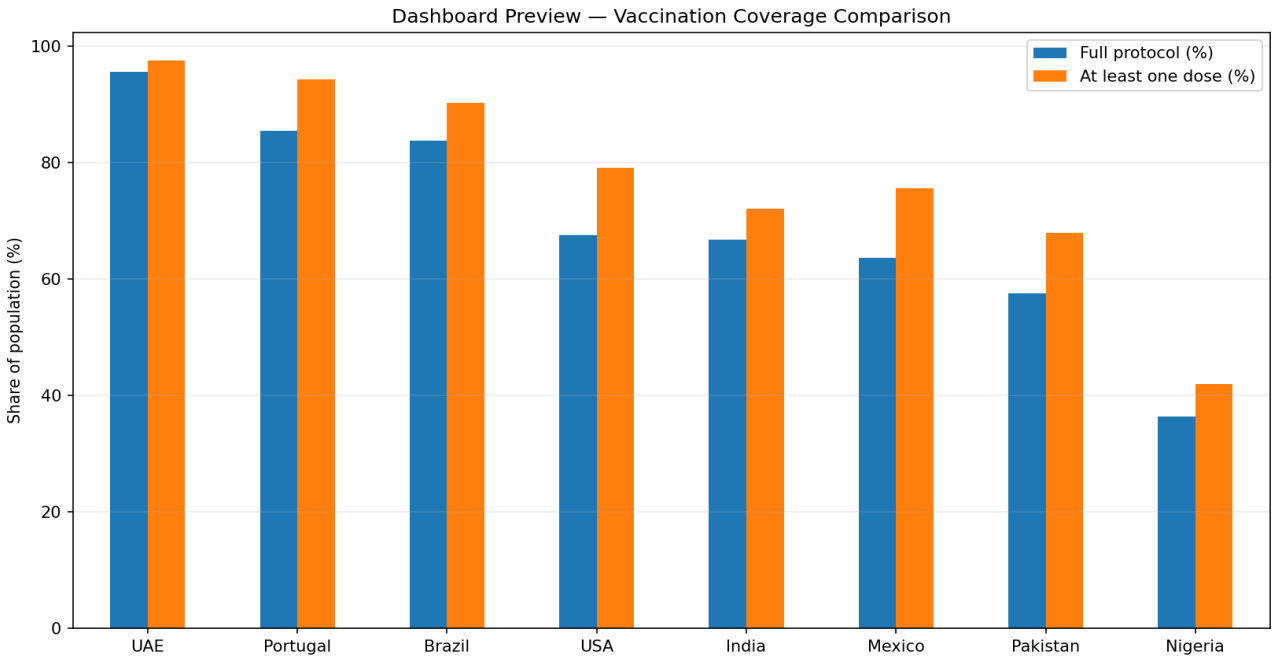
The notebook standardizes dates, converts numerical fields safely, removes duplicate country-date records, sorts observations chronologically, and preserves missing observations as missing. It does not silently convert missing values to zero.

Why missing values matter

A missing observation means the source did not provide a usable value at that time. Replacing it with zero would create artificial drops in cases, deaths or vaccination coverage. This is particularly important for international datasets where reporting systems differ.

Validation checklist

Check	Expected outcome
Country/date uniqueness	One analytical record per country-date after deduplication.
Date ordering	Chronological order within every country.
Population	Positive values before per-million calculations.
Cumulative cases/deaths	Non-negative and interpreted as reported cumulative counts.
Rates	Calculated only where the denominator is positive.
Doubling time	Shown only during positive growth; otherwise left missing.



4. Derived Metrics

7-day rolling average

Daily COVID-19 reports can contain weekday effects, reporting backlogs and administrative corrections. A 7-day rolling average reduces this short-term noise and makes broad waves easier to compare.

Population-normalized rate

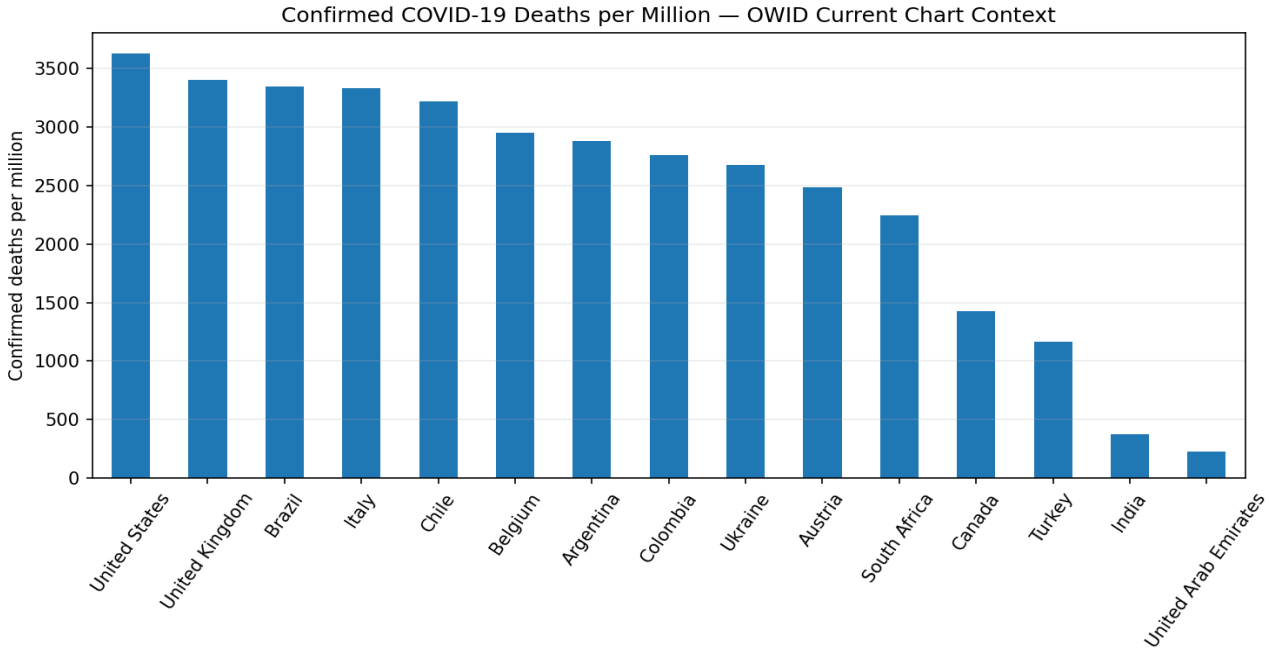
Cases per million = confirmed cases / population × 1,000,000. The same approach is used for deaths. This prevents a large country from automatically appearing more affected simply because it has more residents.

Case fatality rate

CFR = confirmed deaths / confirmed cases × 100. In this project it is explicitly called a crude reported-case fatality ratio because the denominator contains detected cases rather than all infections. Testing intensity and reporting delays can change the ratio.

Doubling time

Doubling time is estimated from seven-day cumulative-case growth. It is reported only when the growth factor is above 1; this avoids meaningless or negative values when cases are flat or falling.



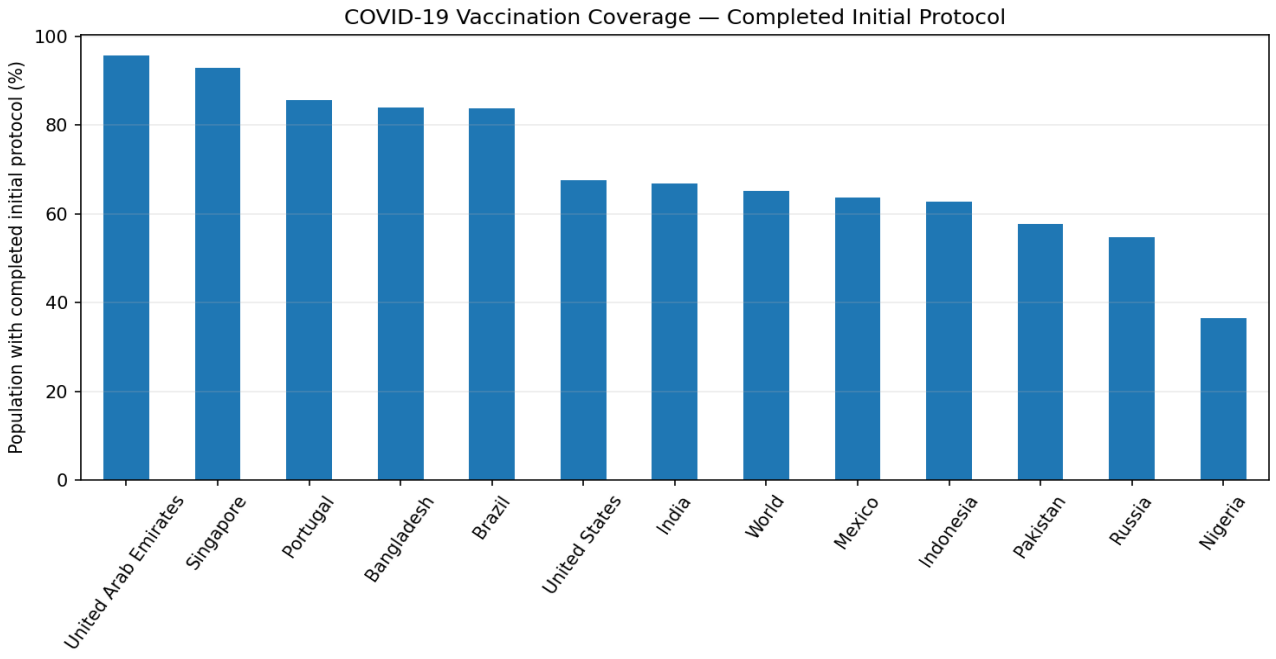
5. Vaccination Rollout Findings

The OWID vaccination snapshot provides a clear example of why percentage coverage is more useful than raw dose counts for cross-country comparison.

Country	≥1 dose	Completed initial protocol
United Arab Emirates	97.5%	95.6%
Portugal	94.3%	85.5%
Brazil	90.2%	83.8%
United States	79.1%	67.5%
India	72.1%	66.8%
Mexico	75.6%	63.6%
Pakistan	67.9%	57.6%
Nigeria	42.0%	36.4%

In the 12 Aug 2024 snapshot, at-least-one-dose coverage ranged from 42.0% in Nigeria to 97.7% in Cuba among the selected preview countries. Completed initial-protocol coverage ranged from 36.4% in Nigeria to 95.6% in the United Arab Emirates among the countries with values shown.

India had 72.1% with at least one dose and 66.8% completing the initial protocol. The United States had 79.1% and 67.5%, respectively. Brazil had 90.2% and 83.8%, while Portugal had 94.3% and 85.5%.



6. Cases, Deaths and Waves

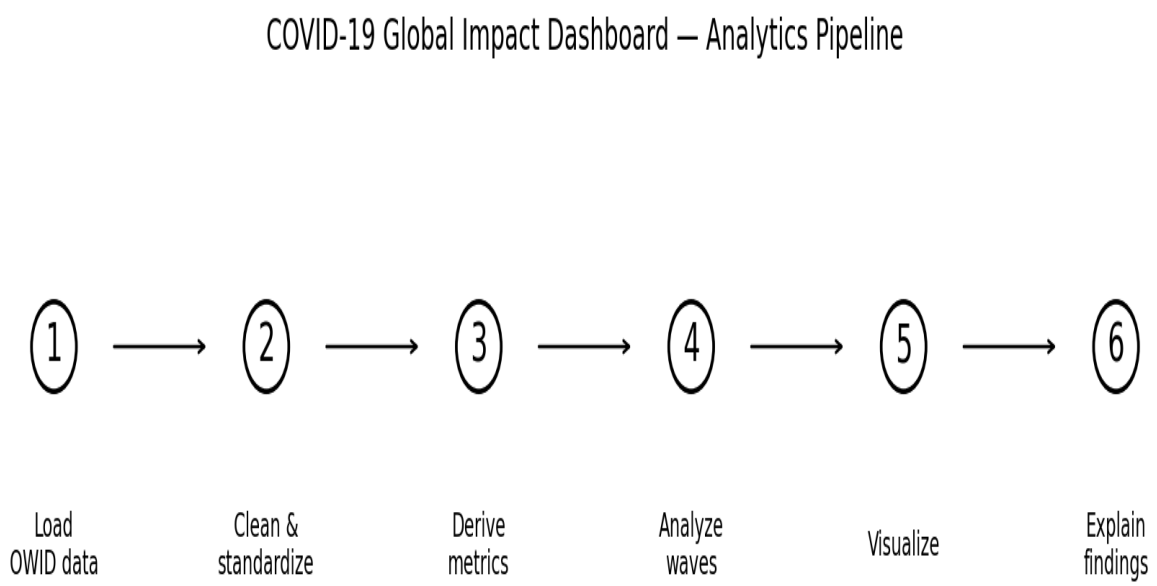
The notebook is designed to identify waves from the smoothed daily case series rather than raw daily observations. A peak is treated as a local maximum in the 7-day rolling average, with a minimum spacing rule to avoid labeling every small reporting fluctuation as a separate wave.

Recommended interpretation

For each country, compare peak date, peak magnitude per million, and the duration of elevated activity. This gives a more useful cross-country comparison than looking only at cumulative totals.

Why waves differ

Observed differences can reflect variant timing, population behavior, public-health measures, vaccination timing, prior exposure, healthcare capacity, testing and reporting practices. The dashboard therefore presents descriptive patterns rather than assigning a single cause.



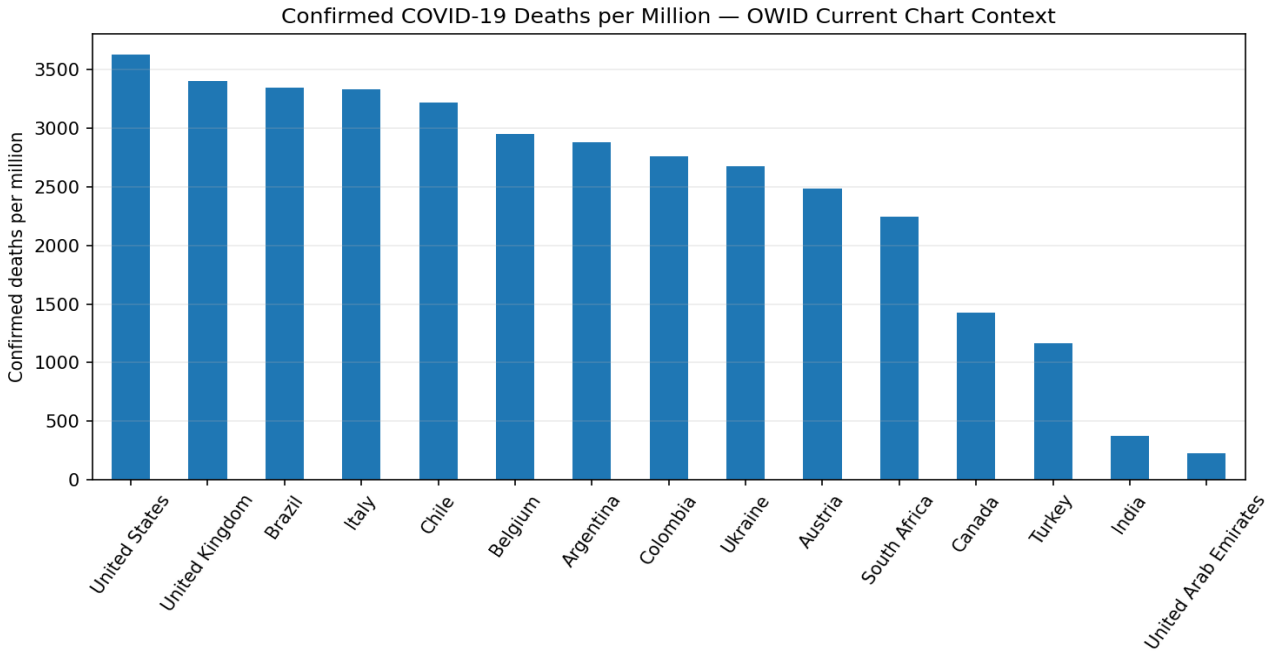
The interactive HTML dashboard also plots cumulative cases per million, cumulative deaths per million and vaccination coverage over time for the selected country.

7. Mortality Context

OWID's current chart context (19 Jul 2026) reports confirmed COVID-19 deaths per million for many countries. The preview values below are included to demonstrate population-normalized mortality comparison; they are not combined with the Aug 2024 vaccination snapshot as if both came from the same date.

Country	Confirmed deaths / million	Chart context
United States	3,626.8	19 Jul 2026
United Kingdom	3,404.4	19 Jul 2026
Brazil	3,347.7	19 Jul 2026
Italy	3,329.9	19 Jul 2026
Chile	3,216.5	19 Jul 2026
Belgium	2,949.6	19 Jul 2026
Argentina	2,881.7	19 Jul 2026
Colombia	2,760.4	19 Jul 2026
Ukraine	2,678.2	19 Jul 2026
Austria	2,485.9	19 Jul 2026
South Africa	2,245.0	19 Jul 2026
Canada	1,424.0	19 Jul 2026

For example, the current OWID chart context shows 3,626.8 confirmed deaths per million for the United States, 3,404.4 for the United Kingdom, 3,347.7 for Brazil, 3,329.9 for Italy and 374.5 for India. These values describe reported confirmed mortality, not the complete true mortality burden.



8. Vaccination and Mortality — How to Read the Relationship

The assignment suggests asking whether vaccination rollout speed correlated with lower death rates. The correct portfolio treatment is to make the relationship visible while explicitly avoiding causal overclaiming.

Recommended analytical design

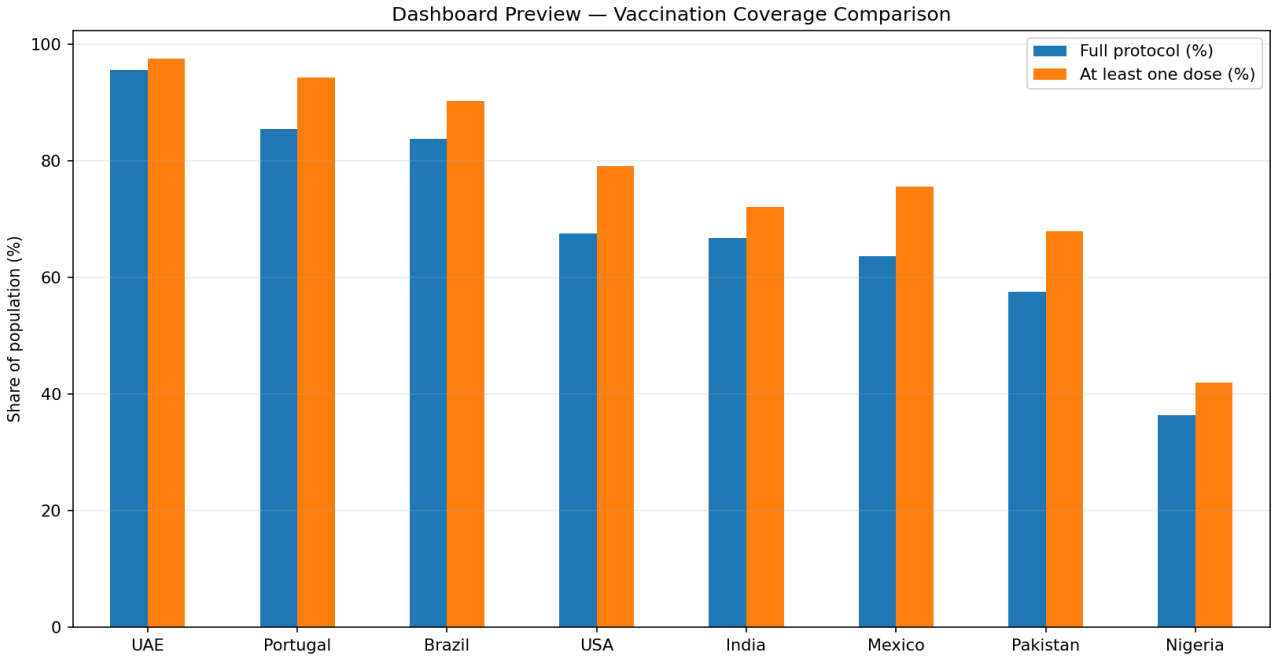
1) choose a synchronized time window; 2) calculate vaccination coverage at defined dates; 3) compare subsequent confirmed-death rates per million; 4) use a scatter plot and time-lagged comparisons; 5) report the correlation coefficient as a descriptive statistic only; 6) discuss major confounders.

Important confounders

Age distribution, prior infection, healthcare capacity, variant circulation, timing of vaccination, booster uptake, testing, reporting practices and policy differences can all influence observed case and death patterns.

What the current preview supports

The preview demonstrates large international differences in vaccination coverage and confirmed mortality. It does not, by itself, establish that vaccination coverage caused those mortality differences. The notebook's synchronized time-series workflow should be used for the final statistical comparison.



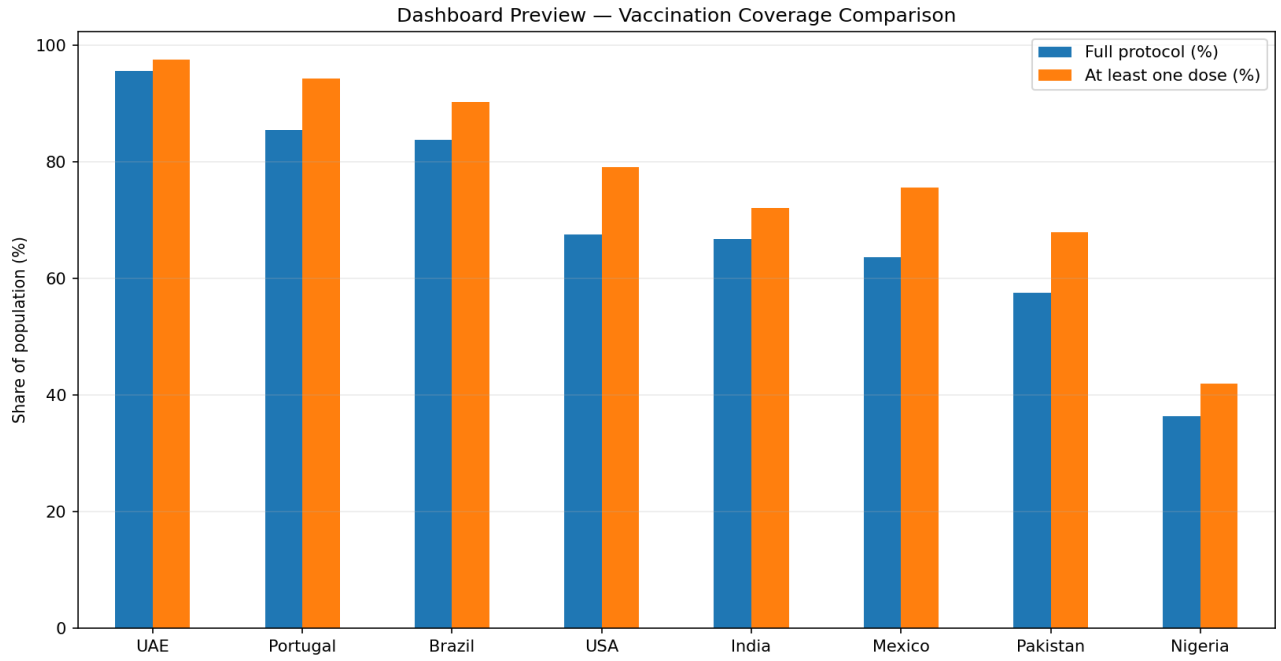
9. Dashboard and User Experience

The interactive HTML dashboard is designed around a simple non-technical workflow: choose a country, review KPI cards, then inspect cumulative cases, cumulative deaths and vaccination rollout over time.

Dashboard elements

Element	Purpose
Country selector	Switch the analysis context without changing the code.
Cases / million KPI	Quick population-normalized cumulative case context.
Deaths / million KPI	Quick population-normalized confirmed mortality context.
Vaccination KPIs	Show at-least-one-dose and completed-protocol coverage.
Case trend	View long-run cumulative reported cases per million.
Death trend	View long-run cumulative confirmed deaths per million.
Vaccination trend	Compare rollout indicators over time.

The dashboard intentionally keeps the visual hierarchy simple: KPIs first, trends second, interpretation last. This makes the project suitable for a portfolio and understandable to a non-technical viewer.



10. Conclusion and Submission Notes

The project meets the requested analytics-pipeline structure: public data acquisition, cleaning, missing-value handling, population normalization, derived metrics, wave analysis, vaccination analysis, visual communication and a dashboard-oriented presentation.

Main portfolio takeaways

• Cross-country comparisons should use per-million metrics rather than raw totals.
• Seven-day rolling averages are more interpretable than noisy daily reporting values.
• Vaccination coverage varied substantially across countries in the OWID Aug 2024 snapshot.
• Confirmed deaths are not equivalent to the full true COVID-19 mortality burden.
• A vaccination/death relationship should be presented as descriptive association unless a stronger causal design is used.

Recommended submission order

1. Public GitHub repository with the notebook, README, outputs and report.
2. Submit the repository URL through the internship portal.
3. Share the completed project on LinkedIn with 2–3 clean output screenshots.

Attribution

Our World in Data and its underlying providers should be credited according to the source metadata and license terms. The README contains the source and documentation links.

End of report — run the notebook before final submission to refresh the exact time-series outputs from the current OWID dataset.